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$$H_3PO_4 + 12\,(NH_4)_2\,MoO_4 + 21\,HNO_3 \xrightarrow{\text{heat}} (NH_4)_3\,PO_4.12MoO_3 + 21NH_4\,NO_3 + 12\,HNO_3$$

The precipitate of ammonium phosphomolybdate thus formed is filtered washed, dried and weighed.

In an alternative method, the phosphoric acid is precipitated as magnesium-ammonium phosphate by adding magnesia mixture (a mixture containing $MgCl_2$, NH_4Cl and ammonia) This ppt is washed, dried and ignited to get magnesium pyrophosphate which is washed, dried a weighed. The following are the reaction that takes place.

By knowing the mass of the organic compound and the mass of ammonium phosphomolybdate or magnesium pyrophosphate formed, the percentage of P is calculated.

Mass of organic compound is wg

Weight of ammonium

phosphomolybdate = x g

Weight of magnesium pyrophosphate = y g

Mole mass of $(NH_4)_3PO_4.12MoO_3$ is =1877g

[3 x(14 + 4) + 31 +4(16)] + 12 (96+3x16)

Molar mass of $Mg_3P_2O_7$ is 222 g

(2x24) + (31x2) + (7x16)

1877g of $(NH_4)_3PO_4.12MoO_3$ contains 31g of P

Xg of $(NH_4)_3PO_4.12\ MoO_3$ in w g of organic compound contains $\frac{31}{1877}\times\frac{x}{w}$ of phosphorous

Percentage of Phosphorous $= \frac{31}{1877}\times\frac{x}{w}\times 100$

(or)

227 of $Mg_2P_2O_7$ contains 62 g of P

Y g of of $Mg_2P_2O_7$ in w g of Organic compound contains $\frac{62}{222}\times\frac{y}{w}$ of P.

Percentage Phosphorous $= \frac{62}{222}\times\frac{y}{w}\times 100$

Example 4: 0.24 g of organic compound containing phosphorous gave 0.66 g of $Mg_2P_2O_7$ by the usual analysis. Calculate the percentage of phosphorous in the compound

Weight of an organic compound = 0.24 g

Weight of $Mg_2P_2O_7$ = 0.66 g

222 g of $Mg_2P_2O_7$ contains 62 g of P

0.66 g contains $\frac{62}{222}\times\frac{0.66}{0.24}$

Percentage of P $\frac{62}{222}\times\frac{0.66}{0.24}\times 100 = 76.80\ \%$

Evaluate Yourself

9) 0.33 g of an organic compound containing phosphorous gave 0.397 g of $Mg_2P_2O_7$ by the analysis. Calculate the percentage of P in the compound (Ans: 23.21) (MFW of $Mg_2P_2O_7$ is 222 P = 31)

Estimation of nitrogen: There are two methods for the estimation of nitrogen in an organic compound. They are 1. Dumas method 2. Kjeldahls method

1. Dumas method:

This method is based upon the fact that nitrogenous compound when heated with cupric oxide in an atmosphere of CO_2 yields free nitrogen. Thus